With this, Darwinism took on a new lease of life. Neo-Darwinism, as we may call the modern theory of gradual transformation operated by natural selection acting on a Mendelian genetic outfit of self-reproducing and self-varying genes, is fully accepted by the great majority of students of evolution. Darwin would have rejoiced to see how, even in the light of our enormously increased knowledge, it (and it alone) can account for the varied and often puzzling facts of evolution—the different types and degrees of adaptation; the (geologically speaking) rapid transformation of some types side by side with the unchanged persistence of others; the coexistence of lower and higher forms; the succession of constantly more improved types; extinction ; the facts of geographical distribution; the evolution of insect societies based on elaborate instincts. Today, a century after the publication of the *Origin*, Darwin's great discovery, the universal principle of natural selection, is firmly and finally established as the sole agency of major evolutionary change.

In the *Origin*, Darwin had already adumbrated many ideas which he or others afterward followed out in more detail. Thus he mentions the existence of plants with two or three types of flower, like primroses, and explains it as a means of avoiding self-fertilization and inbreeding, a subject which he amplified in two books published in 1876 and 1877. He explained how psychological characters like instincts and emotions could and would be evolved by natural means, though it was not until 1872 that, with his remarkable book *The Expression of the Emotions in Man and Animals*, he initiated the comparative study of animal behaviour, or ethology as it is now called. By his constant emphasis on the complicated web of interrelations between different organisms in a given habitat or region he laid the foundations for the new and flourishing branch of science now known as ecology.

In the *Origin* he briefly outlines his theory of sexual selection, which he later (in 1871) developed at great length in his big book *The Descent of Man.* He regarded sexual selection as a subsidiary mechanism of evolution, needed to account for the development of male secondary sexual characters, notably weapons like stags' antlers or exaggerated plumes and conspicuous displays like those of many male birds. Though this theory has been bitterly attacked (often by those without adequate knowledge of the facts), and has had to be modified in various particulars, it provides another example of Darwin's originality and insight. He correctly deduced that such characters, though irrelevant in the general struggle for existence or in competition with other species, would be of advantage in what he called the struggle for reproduction, which must exist between differently endowed males. Selection